# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

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Annexure A

**CONCEPT MAPPING**

*Code of Conduct:*

*I* Palashalika S.V *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

| Criteria | Weightage (%) | Q1 10marks | Q2 10marks | Q3 10marks |
|---|---|---|---|---|
| Concept Identification | 20% | 2 | 2 | 1 |
| Concept Mapping Structure | 25% | 2 | 2 | 1 |
| Linking & Relationships | 20% | 1 | 1 | 1 |
| Application & Analysis | 20% | 1 | 1 | 1 |
| Explanation & Clarity | 15% | 1 | 1 | 0 |

**Total Marks** 18 **/30**

***Signature of the Faculty***

**CONCEPT MAPPING QUESTIONS**

1. A partially completed concept map is given below:

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

## RUBRICS FOR CALCULATION

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Concept Identification** | 20% | Accurately identifies all relevant OSI layers, concepts, and relationships | Identifies most concepts with minor omissions | Identifies limited concepts; some inaccuracies | Unable to identify key concepts correctly |
| **Concept Mapping** | 25% | Constructs a clear, logical, and well-organized concept | Concept map is mostly clear with | Concept map lacks clarity, | No clear structure or |

| *Criteria* | **Weightage (%)** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
|---|---|---|---|---|---|
| ***Linking & Relationships*** | 20% | Clearly establishes correct relationships between layers, functions, and processes | Relationships are mostly correct with minor errors | Limited or partially correct relationships | Incorrect or missing relationships |
| ***Application & Analysis*** | 20% | Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning | The application is correct but lacks depth in analysis | Basic application with weak explanation | Unable to apply concepts correctly |
| ***Explanation & Clarity*** | 15% | Provides a clear, concise, and well-structured explanation supporting the concept map | Explanation is understandable with minor gaps | Explanation is unclear or incomplete | No clear or meaningful explanation |

Computer Networks.

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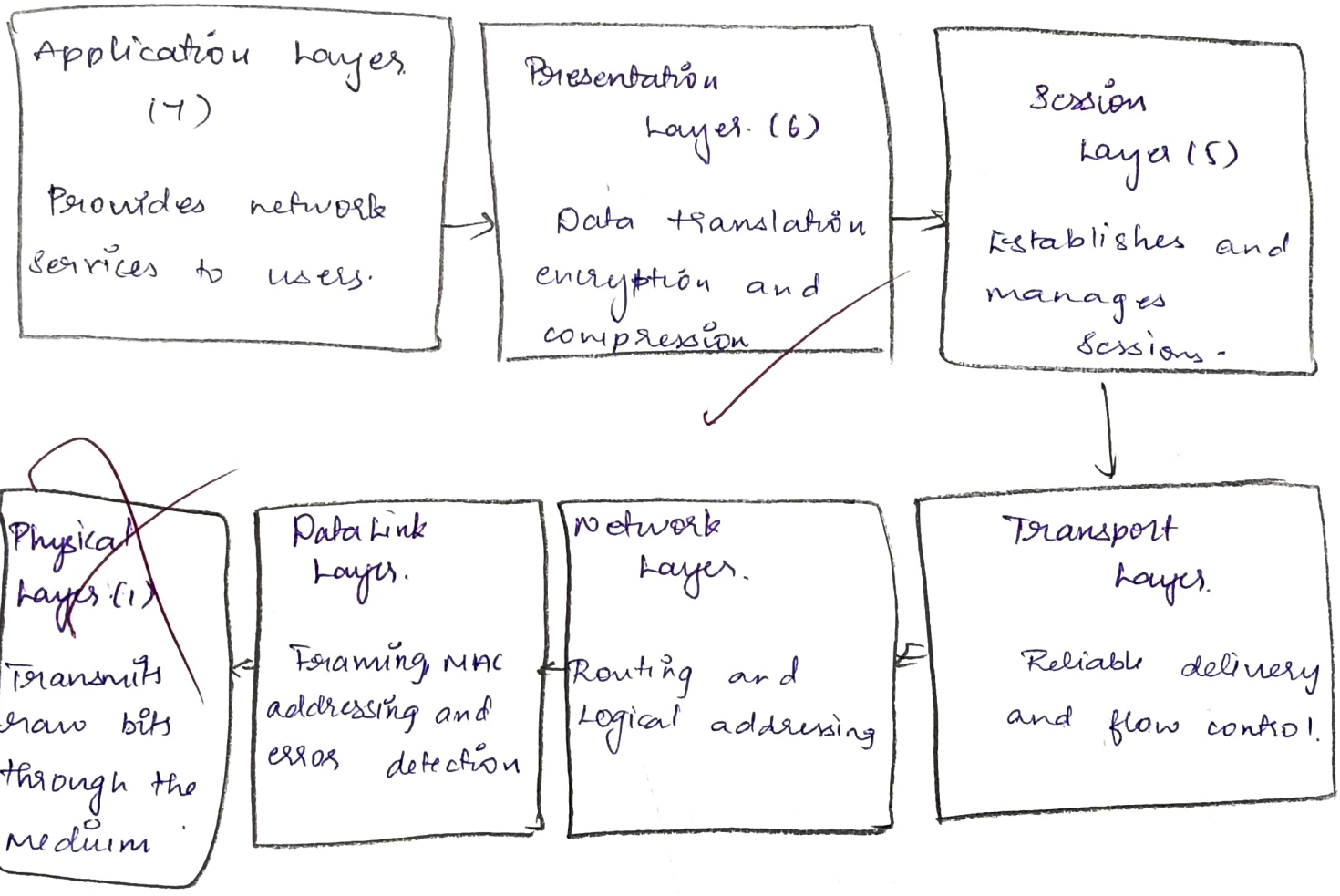
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Faculty Name. R Shankar.

1. Identify the missing OSI layers and explain their functions



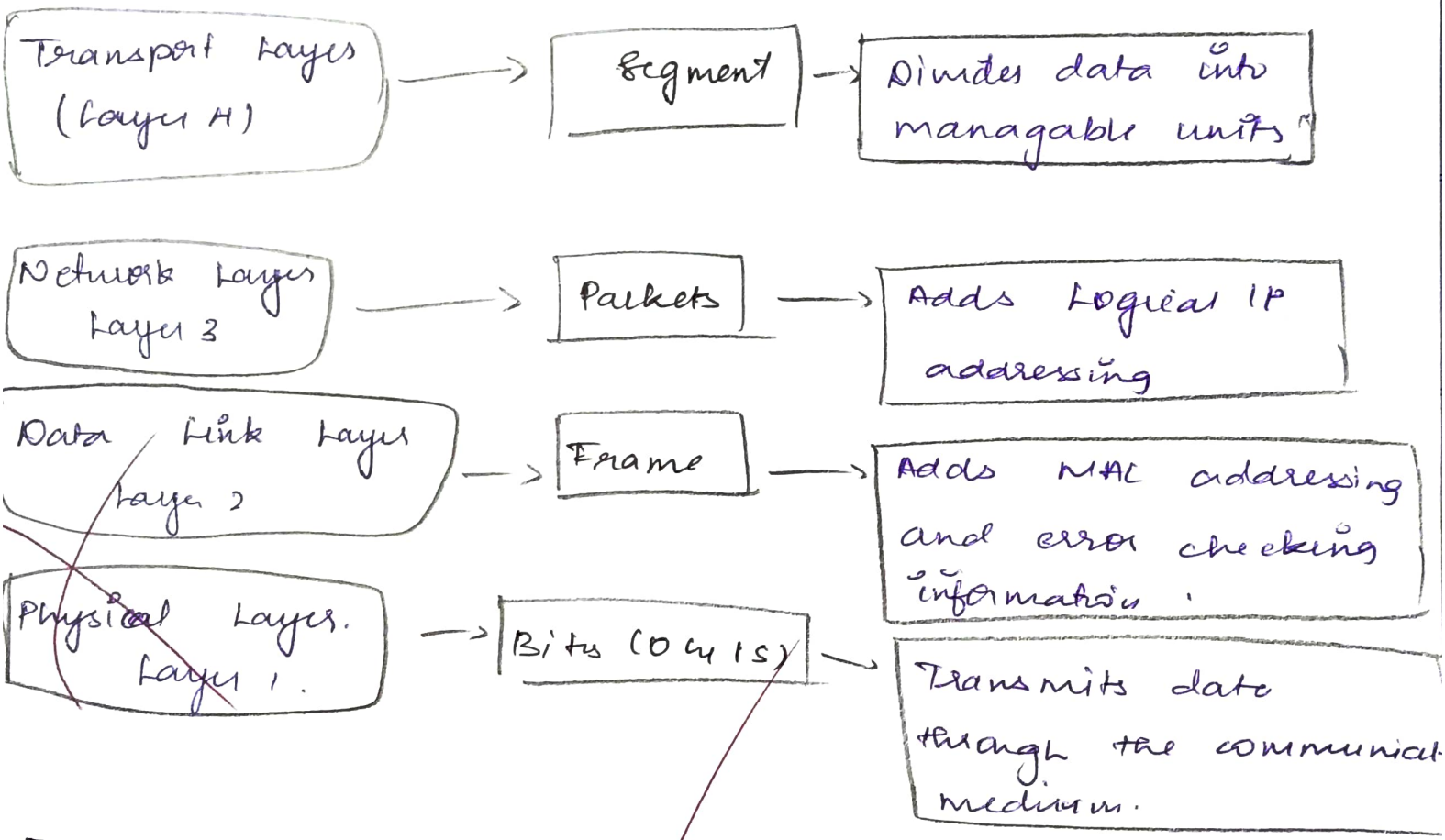

Missing Layers:

Session Layer (5)- Establishes, manages and terminates communication sessions between devices.

Data Link Layer (2)- Transfers data frames from directly connected devices and provides error detection

Construct a concept map linking data with their respective OSI layers and analyze how transformation supports transmission.

Data units and OSI layers.




Transformation supports transmission.

Encapsulation: Data is converted into different data units as it moves down the layers.

Decapsulation: Data is converted back to its original form.

Explain how data link -> MAC address -> Local delivery and network -> IP address -> global delivery ensure complete data delivery.

